

Automatic DC-Offset Calibration for a PSoC-Based Geophone Front End

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Abstract—This paper presents a foreground calibration method for a PSoC-based geophone analog front end in which stage-to-stage accumulation of amplifier and reference offsets can saturate intermediate nodes. The method sequentially addresses four ordered stages – a programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP) – using existing on-chip conversion and filtering resources. Oscilloscope measurements on two independently assembled boards, with reported programmable-PGA settings spanning $\times 1$ – $\times 50$, show ADC-facing LP sample-mean offsets of 0.4–3.9% of the nominal 5-V full-scale span across eight reported board/gain combinations, versus 2.9–45.8% under the fixed-reference baseline. The observed PGA residual is consistent with a deliberately loose, non-optimized acceptance band. Stationary bench operation was observed to complete in approximately 10 s, while the configured per-stage controller timeout windows sum to 57.6 s. A multichannel surface-wave field record with visible first-arrival moveout further demonstrates operation beyond the bench.

Index Terms—Geophone, analog front end, DC-offset calibration, mixed-signal systems, PSoC

I. INTRODUCTION

Multichannel surface-wave methods use distributed vibration measurements to estimate Rayleigh-wave dispersion and infer near-surface mechanical properties [1], [2]. Although their processing is well established, acquisition hardware remains a major source of cost and deployment complexity; compact wireless nodes based on inexpensive geophones and mixed-signal microcontrollers therefore enable denser, reconfigurable arrays [3]. Embedded interfaces already combine amplification, filtering, and programmable gain [4], but achieving low offset through higher-grade analog components raises the cost of every node. This work instead uses resources already present in the microcontroller to correct the operating point of a budget-constrained front end.

The system belongs to a low-cost, reconfigurable Internet-of-Things acquisition platform [5] based on SM-24 moving-coil geophones [6]. Its PSoC 5LP integrates programmable-gain and operational amplifiers, analog filters, voltage DACs, multiplexers, a delta-sigma ADC, a digital filter, DMA, and an Arm Cortex-M3 processor [7]. Time-sharing the ADC and digital filter between acquisition and calibration avoids a separate converter or processor, following digitally assisted analog design [8]. The tradeoff is that the internal analog blocks cannot be independently optimized for offset, drift, and flicker noise, as also reported for a single-chip PSoC potentiostat [9].

The geophone conditioning circuit derives from the cascaded front end of Ma *et al.* [10]. Mapped to the PSoC, accumulated

DC errors consumed operating range at intermediate and final nodes. Because a continuously active servo proved impractical (Section V), the proposed method instead recenters these nodes through finite foreground calibration before acquisition.

On-chip self-calibration is established. PSoC-Stat injects five IDAC levels and uses the ADC response to estimate TIA/ADC gain and offset [9]; Infineon’s AN68403 measures a grounded PGA/ADC chain under an internal VDAC and stores numerical corrections in EEPROM [11]. Physical trimming has also been demonstrated for a single amplifier through body-bias calibration [12]. Here, instead of correcting only the ADC transfer characteristic or one amplifier, the loop adjusts the physical reference of each trim-accessible analog node within the available DAC range and resolution.

The front end remains DC-coupled because a DC-blocking input capacitor would require a corner well below 1 Hz to preserve the SM-24 low-frequency response. Per-stage trimming addresses offset without abandoning the single-chip implementation or introducing classical autozero/chopper circuitry [13].

The contributions are: 1) a four-VDAC per-stage architecture that reuses the PSoC multiplexer, ADC, and digital filter; 2) a bounded deadband controller that freezes its codes before acquisition; and 3) two-board validation over programmed-PGA settings $\times 1$ – $\times 50$ without manual controller-parameter retuning.

II. PLATFORM DESIGN

The PSoC conditions and digitizes the geophone output in acquisition mode. In calibration mode, its measurement path, digital-filter coefficients, and DAC-control firmware form a temporary mixed-signal loop without changing the sensor input chain.

In the feed-forward signal chain of Fig. 1, an offset introduced at any stage can propagate through downstream gains toward the ADC input. The path is not a strict one-input cascade: BP follows PGA, whereas SUM combines a direct PGA branch with the BP branch before LP. The calibration therefore exposes and trims each stage separately rather than correcting only the final ADC value.

An analog multiplexer connects one stage output at a time to the shared 18-bit delta-sigma ADC, which serves as the error sensor; software mutual exclusion prevents simultaneous connections. A switched capacitor behind the selected multiplexer MOSFET combines with its on-resistance for limited RC

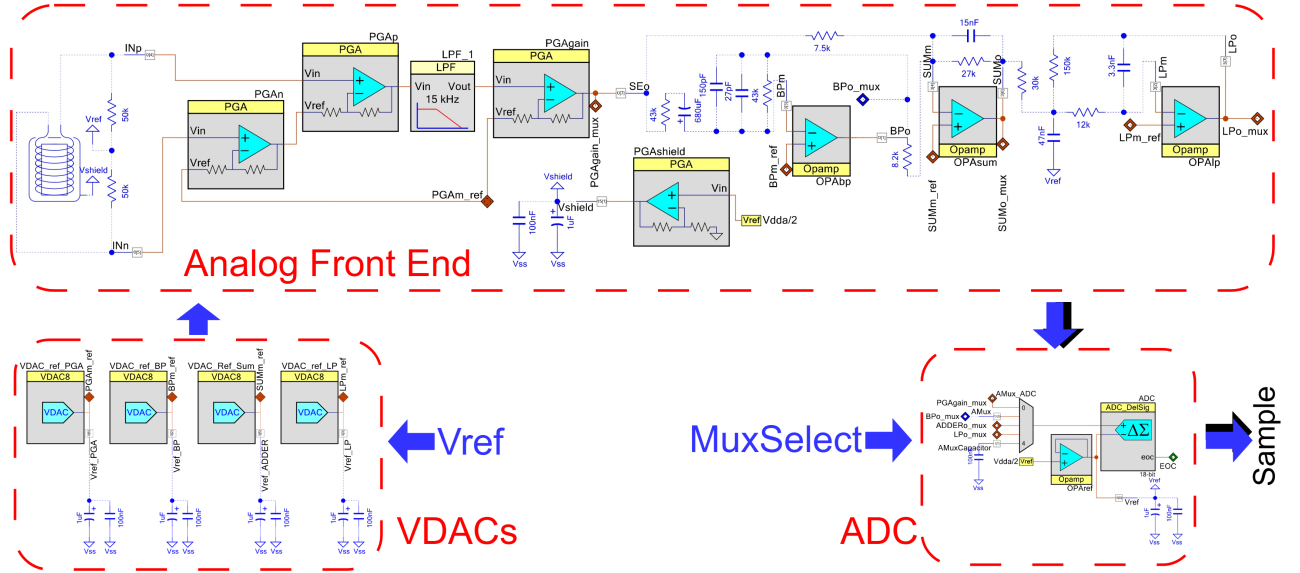


Fig. 1. Analog signal chain: four independent VDACS trim the programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP), with a shared ADC/multiplexer measurement path.

anti-aliasing and isolates the amplifier from a direct capacitive load.

For the i th measured node, let y_i^{DC} denote its slowly varying voltage deviation from the intended stage reference, with polarity matching the differential ADC coordinates. It can be represented as

$$y_i^{\text{DC}} = \sum_{j < i} A_{ij} y_j^{\text{DC}} + b_i + K_i u_i, \quad (1)$$

where A_{ij} is the DC coupling from an upstream node j (zero when no such path exists), b_i collects amplifier and reference errors in voltage coordinates, u_i is the VDACS voltage, and K_i is the signed DC gain from that VDACS to the measured node. The firmware represents u_i by an 8-bit command code c_i , such that $u_i = q_{\text{DAC}} c_i$. The multiple-input form covers both ordinary cascaded stages and nodes such as SUM; the calibration target is $y_i^{\text{DC}} = 0$.

The proposed architecture assigns a dedicated trim input u_i to each trim-accessible node in Eq. 1 and determines the codes sequentially so that every measured y_i^{DC} approaches zero, i.e., its stage reference in the differential ADC coordinates.

With negligible reverse coupling and adequate VDACS authority and range, the lower-triangular model in Eq. 1 permits one forward pass in PGA, BP, SUM, and LP order (Section III).

III. SELF-CALIBRATING GEOPHONE PLATFORM

A. Digital Hardware and Calibration Sequence

At boot, after a gain change, or on a calibration command, the deployment firmware loads stored DAC codes and verifies the resulting offsets (Fig. 2). If the operating point is accepted, it restores acquisition without a full run; otherwise it processes PGA, BP, SUM, and LP sequentially, resetting the FIR history and settling after each multiplexer switch. A watchdog and per-stage sample limit bound the sequence. During acquisition

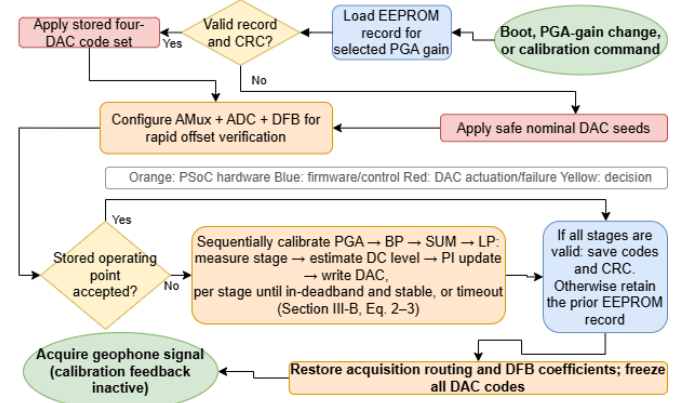


Fig. 2. Foreground calibration flow: stored-code verification, sequential trimming, and acquisition.

the controller is inactive, the multiplexer selects LP, and the VDACS codes remain constant, so no time-varying feedback enters the acquired signal.

The shared digital filter implements the geophone response during acquisition and switches to a 128-tap low-pass FIR DC estimator during calibration (Fig. 3); its history is cleared at each node, and its acquisition coefficients and path are restored afterward. Verified DAC codes can be stored in EEPROM as one record per PGA gain setting containing the gain code, valid-stage mask, four DAC codes, version, and CRC. A failed run never overwrites a valid record; its bounded best codes may be held for the current sequence, but they are not promoted to a verified EEPROM record.

B. Control Law and Calibration Parameters

For each stage, the ADC reports the node voltage relative to V_{ref} . Thus, $\hat{y}_i[n] = 0$ denotes a physical node at V_{ref} , not at 0 V. The filtered differential measurement produces the ADC-

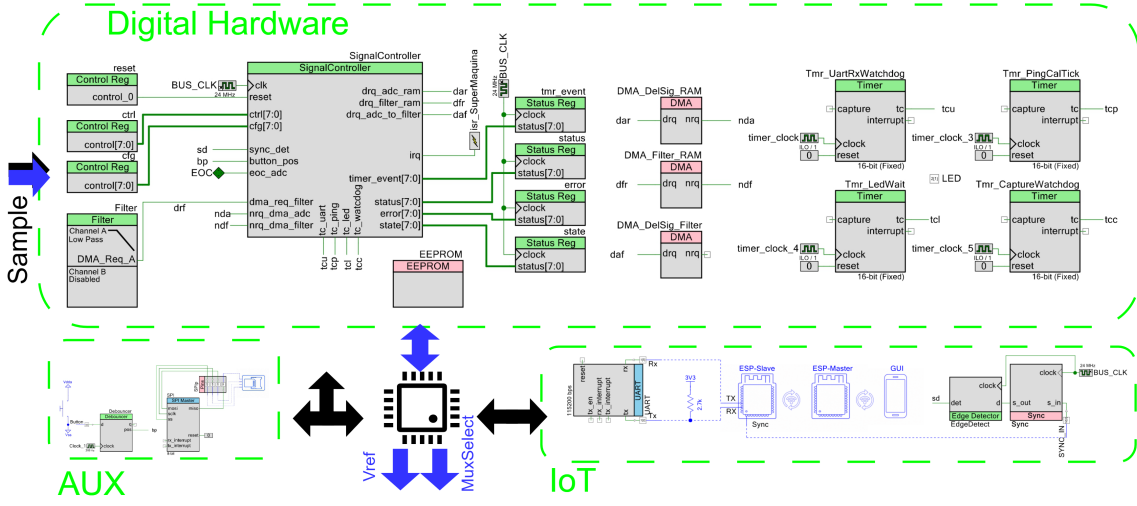


Fig. 3. Digital calibration hardware. The controller routes DMA transfers among the ADC, shared digital filter, and EEPROM and reconfigures the analog multiplexer. Black and blue arrows indicate acquisition and calibration paths, respectively.

domain error $e_i[n] = r_i - \hat{y}_i[n]$, with $r_i = 0$. The error is first mapped to the 8-bit DAC domain:

$$\begin{aligned} \varepsilon_i[n] &= \text{round}\left(e_i[n] \frac{q_{\text{ADC}}}{q_{\text{DAC}}}\right) \\ &= \text{round}\left(\frac{e_i[n] V_{\text{ADC,span}} N_{\text{DAC}}}{N_{\text{ADC}} V_{\text{DAC,span}}}\right), \end{aligned} \quad (2)$$

where $q_x = V_{x,\text{span}}/N_x$ is the step size and N_x is the number of quantization levels, not the maximum code. Here $N_{\text{ADC}} = 2^{18}$, $N_{\text{DAC}} = 2^8$, $V_{\text{ADC,span}} = 5000$ mV, and $V_{\text{DAC,span}} = 4096$ mV. Thus, $q_{\text{ADC}} = 5000/2^{18}$ mV/count and $q_{\text{DAC}} = 4096/2^8 = 16$ mV/code.

The rounding in Eq. 2 makes $\varepsilon_i[n]$ an integer error in equivalent DAC codes. If the ideal operating point lies between two realizable codes, an overly narrow acceptance band can make the controller alternate indefinitely between adjacent DAC codes. An ideal nearest-code argument requires only half an actuator step, but the implementation adopts the more conservative one-step floor

$$\Delta_{i,\text{min}}^{\text{impl}} = \max(1, \lceil |K_i| \rceil), \quad (3) \quad \bar{\varepsilon}_i[n] = \begin{cases} 0, & |\varepsilon_i[n]| \leq \Delta_i, \\ \varepsilon_i[n], & \text{otherwise,} \end{cases} \quad (4)$$

Inside the band, Eq. 4 suppresses corrective integration; lock and refinement still govern convergence. Used values satisfy $\Delta_i \geq \Delta_{i,\text{min}}^{\text{impl}}$ and are listed beside the adopted floors in Table I.

The architecture requires only a measurable stage output and sufficient trim range. The controller is written in positional PI form,

$$c_i^*[n] = c_{i,0} + s_i \left[\frac{k_{p,i}}{|K_i|} \bar{\varepsilon}_i[n] + \frac{k_{i,i}}{|K_i|} \sum_{q=0}^n \bar{\varepsilon}_i[q] \right], \quad (5)$$

Here $c_{i,0}$ is a feedforward code near the expected operating point and, under the adopted error convention, $s_i = \text{sgn}(K_i)$. The gain magnitudes in Table I follow from the topology and resistor ratios; the gain-2 two-op-amp input of [14], followed by the programmable-gain block (Fig. 1), gives nominal

TABLE I
IMPLEMENTED PER-STAGE PARAMETERS. THE ONE-STEP FLOORS AND USED DEADBANDS ARE IN EQUIVALENT DAC CODES; TIMING ENTRIES ARE IN SAMPLES AT 2604 HZ.

Stage	$ K_i $	$\Delta_{i,\text{min}}^{\text{impl}}$	Δ_i used	$k_{i,i}$	Settle	Lock	Refine	Timeout (s)
PGA	1	1	18	3×10^{-4}	512	512	1024	11.5
BP	1	1	3	5×10^{-4}	512	1024	1024	17.3
SUM	7.89	8	10	5×10^{-4}	512	512	1024	11.5
LP	6	6	8	5×10^{-4}	1024	1024	2048	17.3

differential signal gain $2 \text{Gain}_{\text{PGA}}$, whereas its DC VDAC-to-node gain is $K_{\text{PGA}} = 1$. With $k_{p,i} = 0$, Eq. 5 reduces to a pure integral law.

The command is clamped to the VDAC range and slew-limited to one code per sample, with conditional integration for anti-windup. Convergence requires a fixed number of samples in the same quantized bucket; a final adjacent-code trial is retained only when it lowers the filtered error. At timeout, the best code found is held, the stage is flagged not converged, and the sequence advances so downstream behavior can still be evaluated. The new code set is stored only if every controllable stage succeeds; otherwise the prior EEPROM record is retained and the run remains flagged as failed.

Applications requiring finer trim resolution can use a PSoC IDAC/resistor or buffered topology [15] without changing the measure–estimate–correct sequence.

IV. EXPERIMENTAL VALIDATION

A. Prototype and Comparison Conditions

Two independently assembled SM-24 conditioning prototypes (Board A and Board B), each with four VDAC references, were compared under *Default*, the same fixed four-code baseline selected on Board B at $\times 50$ and reused at every tested gain, and *Proposed*, four-stage calibration followed by frozen-code acquisition. Default is fixed, not uncalibrated. For repeatability, the measurement firmware

TABLE II
RESIDUAL DC OFFSET AT EACH STAGE AS % OF THE NOMINAL 5-V ADC SPAN, REPORTED IN SEPARATE DEFAULT AND PROPOSED ROWS AS MEAN $\pm$ SAMPLE SD ($N=10$ REPEATED RESTARTS PER REPORTED CELL; LOWER IS BETTER).

PGA setting	Condition	Board A				Board B			
		PGA	BP	SUM	LP	PGA	BP	SUM	LP
$\times 1$	Default	3.5 ± 0.2	1.5 ± 2.2	$21.8 \pm 1.1^\dagger$	$45.8 \pm 3.1^\dagger$	—	—	—	—
	Proposed	3.4 ± 0.3	1.2 ± 0.2	0.8 ± 0.2	3.2 ± 0.9	—	—	—	—
$\times 4$	Default	2.4 ± 0.1	4.9 ± 0.3	6.7 ± 1.0	$42.6 \pm 4.2^\dagger$	—	—	—	—
	Proposed	2.1 ± 0.3	1.0 ± 0.2	1.0 ± 0.3	3.9 ± 0.8	—	—	—	—
$\times 16$	Default	2.4 ± 0.1	3.8 ± 0.9	4.9 ± 1.5	$29.6 \pm 6.8^\dagger$	2.1 ± 0.1	0.66 ± 0.02	4.5 ± 0.8	3.0 ± 0.1
	Proposed	1.9 ± 0.2	1.5 ± 1.5	1.4 ± 1.5	2.3 ± 1.4	1.7 ± 0.1	0.66 ± 0.02	0.6 ± 0.5	0.7 ± 0.1
$\times 32$	Default	1.0 ± 0.2	7.0 ± 0.4	7.4 ± 0.3	$12.4 \pm 1.3^\dagger$	1.0 ± 0.2	0.66 ± 0.01	8.0 ± 0.8	2.9 ± 0.1
	Proposed	0.5 ± 0.2	1.0 ± 0.5	0.6 ± 0.2	1.7 ± 0.7	0.5 ± 0.2	0.66 ± 0.01	1.2 ± 0.3	0.4 ± 0.2
$\times 50$	Default	1.2 ± 0.3	1.0 ± 0.7	0.4 ± 0.2	4.1 ± 2.1	2.5 ± 0.1	0.65 ± 0.01	0.9 ± 0.6	2.9 ± 0.1
	Proposed	1.0 ± 0.5	0.9 ± 0.5	$0.8 \pm 0.6^\ddagger$	1.2 ± 0.4	1.8 ± 0.3	0.65 ± 0.02	0.9 ± 0.6	0.7 ± 0.2

† Default $\geq 10\%$ FS. ‡ The marked Proposed mean is larger than Default; equality after rounding alone is not marked (Section V-A). Dashes denote cells without an $N=10$ comparison; the separate Board B $\times 4$ three-run diagnostic is excluded.

disabled the post-calibration EEPROM write shown in Fig. 2, so every restart reloaded that baseline rather than previously calibrated codes. With the geophone stationary, each Proposed run followed the normal verification/correction sequence; the $N=10$ observations per reported cell are separate baseline-seeded paired restarts. The normal controller made no update on Board B $\times 1$ or $\times 4$. A separate Board B $\times 4$ diagnostic set all $\Delta_i = 1$ to force action for three before/after pairs; it is excluded from Table II because its bands and sample count differ.

B. Residual DC-Offset Measurement and Results

Offsets are normalized to the nominal 5-V span; calibration accepts a band and requires every measured node to remain linear and unsaturated. Section V-A discusses the relaxed PGA criterion.

At each reported board/gain setting, four channels of a Tektronix TDS 2004B 8-bit, 60 MHz oscilloscope recorded V_{mean} at PGA, BP, SUM, and LP over a 100 ms post-settling window. All probe reference leads were tied to the physical $V_{\text{ref}} = V_{\text{DDA}}/2$ node; V_{DDA} was supplied by the PC USB port and was typically slightly below 5 V. Each channel therefore measured $V_{\text{stage}} - V_{\text{ref}}$ directly. The PC and oscilloscope shared the same mains branch, but only the oscilloscope had protective earth, making this probe connection the DUT's only protective-earth path. Within each paired restart, Default was recorded from the reapplied baseline before normal verification/correction and Proposed afterward; the order was not randomized, so slow drift remains a possible confound.

For each run and stage, $100|V_{\text{mean}}|/5$ V was calculated, and Table II reports the mean and sample SD over ten paired restarts. The SD characterizes run-to-run repeatability under the same bench conditions, not unit-to-unit reproducibility. No independent oscilloscope uncertainty budget was established; the displayed digits preserve the calculations but do not imply sub-millivolt absolute accuracy.

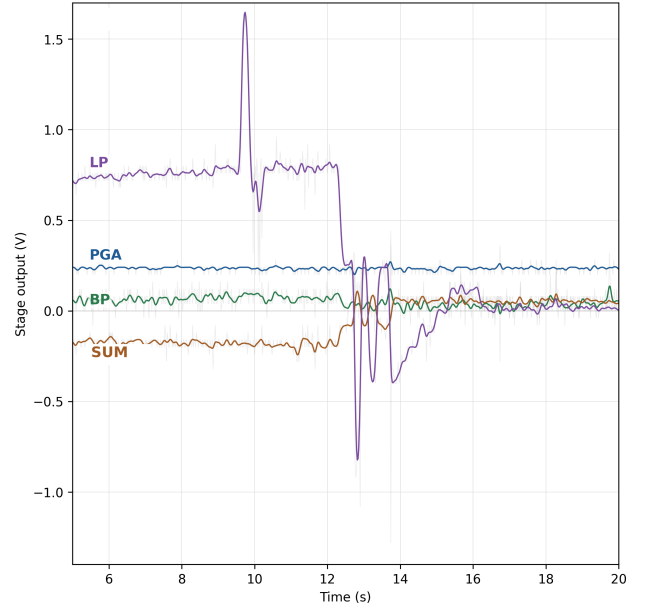


Fig. 4. Stage deviations relative to V_{ref} during calibration at setting $\times 16$ (total differential gain $\times 32$). Thin gray traces are recorded samples; colored curves are low-pass-smoothed visual guides. See Section V-A for interpretation.

Controller acceptance uses the filtered internal ADC estimate acquired during calibration, whereas Table II uses a subsequent oscilloscope record. The observations were asynchronous, and final ADC errors and acceptance flags were not logged with the scope window. Consequently, the scope residuals do not directly verify Δ_i and may exceed its nominal voltage equivalent; the effects of estimator bandwidth and elapsed time were not separated.

Across the eight reported board $\times$ gain cells, the LP sample mean is 0.4–3.9%FS after calibration versus 2.9–45.8%FS under Default; individual Proposed observations span 0.1–5.2%FS (Table II). Within the tested cells, calibration therefore yields a substantially smaller ADC-facing offset without manual

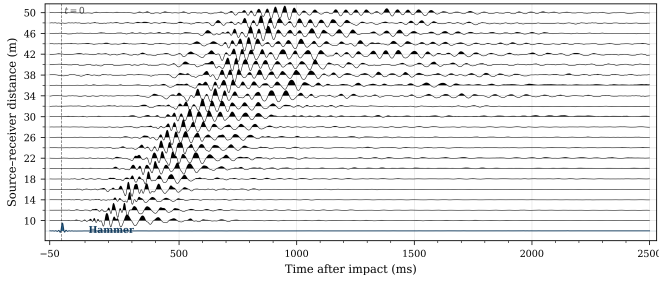


Fig. 5. Wigggle-trace field record acquired with the calibrated front end. The hammer trigger and first-arrival moveout are visible across the 10–50 m line.

retuning.

C. Field Use of the Calibrated Front End

Separate from the bench offset test, Fig. 5 shows wigggle traces from a field surface-wave record over a 10–50 m line. Calibration was initiated after each geophone was planted and stationary. The only observed timeout condition occurred during field handling when a board was powered while its geophone was still moving: the large AC transient drove the chain into saturation and violated the quasi-static-input assumption of the DC estimator. First-arrival moveout is resolved across the line; this demonstrates field operation but is not a controlled Proposed-vs-fixed-reference comparison.

V. DISCUSSION

Unlike precision front ends characterized near the sensor’s Johnson-noise limit [16], this method trades component precision for measurement and digital assistance. Its benefit over software subtraction is physical recentering of the analog nodes during acquisition. It covers observable additive DC errors within DAC authority and range, without identifying their source. The comparison evaluates the complete sequence against the fixed baseline, not each trim separately; per-stage ablation remains future work.

A tested continuous DC servo settled impractically slowly at a low corner, interfered with the signal at a higher corner, and restarted after gain changes [17]; the foreground controller instead holds constant DAC outputs during acquisition. Limitations include finite DAC resolution, no noise or distortion correction, and unvalidated convergence under sustained vibration or thermal drift.

A. Deliberately Relaxed PGA Acceptance

PGA post-calibration offset is only modestly smaller than Default. Since $|K_{PGA}| = 1$, one equivalent code is 16 mV and $\Delta_{PGA} = 18$ defines a ± 288 mV band. The 0.24-V level in Fig. 4 lies inside it and is consistent with the intended stop-on-band behavior; a smaller value requires evaluation. In the separate Board B $\times 4$ diagnostic, setting all $\Delta_i = 1$ forced updates and reduced the three external LP residuals from 148/142/157 mV to 34.4/38.5/53.5 mV. Because every stage band changed and only three pairs were collected, this excluded diagnostic does not establish a PGA accuracy floor. Motion-aware estimation and systematic deadband selection remain future work.

The sole reversal, Board A SUM at $\times 50$ (0.4%FS to 0.8%FS), is compatible with band acceptance and does not affect the downstream LP result.

Visible correction in Fig. 4 spans about 7 s. Stationary operation was observed near 10 s but was not timed for every repetition. The four controller-update windows in Table I sum to 57.6 s; adding the separately configured settling and refinement intervals gives an approximately 60.6-s ceiling, with lock acquisition already included in those windows. No laboratory run reached it.

One nominal initialization selected on Board B at $\times 50$ transferred to Board A and the reported gain settings. However, $N=10$ measures only within-cell repeatability, and two boards cannot establish population scaling; more units and cross-temperature trials are required.

VI. CONCLUSION

Using four VDACS and existing PSoC resources, per-stage foreground calibration reduced the ADC-facing LP sample-mean offset from 2.9–45.8%FS to 0.4–3.9%FS across eight cells on two boards, with codes frozen during acquisition. A field record demonstrated operation over 10–50 m. Population, temperature, VDAC-reference noise, controller tuning, ablation, and controlled field validation remain future work.

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